

Moez Dbira

AI Engineer | Agentic AI, LLM Systems & Production ML

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PROFILE

AI Engineer / Data Scientist with approximately five years of professional experience across machine learning, data engineering, and AI systems. Builds LangGraph agent workflows, access-controlled RAG applications, and Python/FastAPI services integrated into production environments. Additional experience includes model evaluation, observability, asynchronous processing, and collaboration with product and engineering teams.

TECHNICAL SKILLS

Agentic AI and LLMs: LangGraph, RAG, vLLM, embeddings, hybrid retrieval, Qdrant, OpenSearch, reranking, LangSmith

Programming and APIs: Python, SQL, R, FastAPI, Pytest, PostgreSQL, Redshift, SQLite, asynchronous processing

Machine Learning and Data Science: Scikit-learn, LightGBM, XGBoost, PyTorch, TensorFlow, Pandas, NumPy, time-series forecasting, NLP, computer vision, reinforcement learning

Data Engineering and MLOps: Airflow, dbt, Great Expectations, Docker, Kubernetes, MLflow, ONNX, Redis, GitHub

Cloud and Big Data: AWS (S3, RDS, Redshift, Lambda, EC2), HDFS, MinIO

Visualization and Observability: Tableau, QuickSight, Grafana, Prometheus, OpenTelemetry, Plotly, Streamlit

PROFESSIONAL EXPERIENCE

Meritis Technologies | Data Engineer / Data Scientist

December 2021 – Present | France

Consulting assignments in data engineering, machine learning, and AI engineering.

AI Data Reliability Agent

- Designed and productionized a data-incident analysis workflow, from FastAPI ingestion of Airflow/dbt and data-quality signals through Redis asynchronous processing and LangGraph orchestration.
- Structured specialist agents for pipeline failures, data quality, freshness, schema drift, and SQL analysis, followed by an RCA aggregation stage with typed outputs.
- Connected Qdrant hybrid retrieval and reranking to incident history and runbooks, added evidence validation, confidence scoring, deterministic fallback, and human review.
- Instrumented the workflow with OpenTelemetry and LangSmith, optimized LLM context through error-window extraction, log deduplication, SQL normalization, and context budgets.

Confluence RAG Assistant | Business Requirements and Jira Stories

- Designed initial and incremental ingestion for an estimated 15,000–30,000-page Confluence knowledge base, indexing metadata and filtering retrieval by access permissions.
- Implemented hybrid vector/BM25 retrieval with OpenSearch and cross-encoder reranking to produce structured, sourced answers linked to authorized Confluence pages.
- Built workflows for generating Business Requirements, Technical Specifications, and Jira stories, with human review before ticket creation, integrated the web assistant with Confluence and Jira through SSO.
- Reduced the initial documentation-search and drafting cycle from several days to approximately 30–60 minutes of assisted drafting and review.

Meritis Technologies | Data Engineer / Data Scientist (continued)

Enterprise Pricing Data Engineering Mission

- Owned the Pricing Data Engine ETL framework and technical specifications supporting pricing governance, contract execution, rebate settlement, and margin analysis.
- Modernized Python/SQL workflows with Airflow and dbt, incremental loads, layered transformations, star-schema data marts, and stage-level quality checks.
- Improved pipeline performance, delay, and data-quality monitoring, defined data-mart requirements with Business Analysts and migration strategies with Cloud Ops and support teams.

iPhone Sales Forecasting

- Built a production batch pipeline generating weekly, eight-week-ahead forecasts for each iPhone model.
- Compared seasonal baselines with Random Forest, XGBoost, LightGBM, and CatBoost using time-based backtesting, implemented weekly refreshes, scheduled or performance-triggered retraining, and drift monitoring.

Gaming App Ads Recommendation

- Designed a matrix-factorization collaborative-filtering model to rank ads by user–ad affinity, evaluated rankings with Precision@K and Recall@K and contributed to online integration.

Earlier Data Science and Data Engineering Internships

2019 – 2021 | Paris, France

- **Generix Group | May–November 2021:** Developed computer-vision prototypes for warehouse operations and applied constrained optimization and reinforcement learning to truck-loading problems.
- **Virtuo Technologies | March–August 2020:** Improved demand-forecasting workflows, automated model retraining, and optimized vehicle allocation using Python and R.
- **Generix Group | July 2019–January 2020:** Built a Python ETL pipeline from Amazon S3 to Redshift, calculated more than 100 KPIs for QuickSight reporting, and applied SARIMAX to warehouse workload forecasting.

EDUCATION

CentraleSupélec

2017 – 2021 | Studies in Artificial Intelligence, Data Science & Engineering

Institut Préparatoire aux Etudes Scientifiques et Techniques (IPEST)

2014 – 2017 | Preparatory studies in Mathematics and Physics

LANGUAGES

English: Professional | **French:** Bilingual | **Arabic:** Native

LEADERSHIP & INTERESTS

- Football, tennis, manga, music, and travel.
- **VP Voyage, Bureau des Internationaux (Bdl), CentraleSupélec:** Organized international student trips and cultural events.
- Co-founded the first student council at Lycée Pilote de Gabès.
- Contributed to the organization of two TEDx events in Gabès.